LESSON 108 (0.5 GROUND and 1.3 DUAL)

Lesson Objective

During this lesson, the applicant will apply the techniques practiced in the training device in the aircraft. The applicant will be introduced to tracking along a road to increase the applicants understanding of wind drift and wind correction that can later be correlated to the traffic pattern and approach and landing. The applicant will demonstrate straight and level flight at various airspeeds and configurations with instructor guidance to enhance their understanding of aircraft controls and the effects power and configuration have on the aircraft state. The applicant will also review previously learned maneuvers for proficiency.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
    - Quiz the student on the readings as needed.
        - Chapter 7 of the Airplane Flying Handbook - Ground Reference Maneuvers:
            - Drift and Ground Track Control.
- Student questions.

References

- Airplane Flying Handbook. (FAA-H-8083-3C)
    - Chapter 02 - Ground Operations.
    - Chapter 03 - Basic Flight Maneuvers.
    - Chapter 06 - Takeoffs and Departure Climbs.
    - Chapter 07 - Ground Reference Maneuvers.
- Warrior Maneuvers Manual.

Instructor guidance

- This is the student's second flight in the actual aircraft. The primary goal is transferring what was practiced in the simulator to the real airplane — manage expectations, as the aircraft will feel different from the AATD in terms of control sensitivity and environmental factors.
- During the ground portion, quiz the student on taxiing procedures from Lesson 107 before heading out to the aircraft. If they cannot recall the before-taxi check, address it on the ground rather than at the hold short line.
- When introducing the communications and navigation units, keep it to what is immediately relevant for the flight. The student does not need to master the Garmin 430 today — focus on frequency changes, volume, squelch, and the transponder. Save deeper navigation functions for later lessons.
- Ensure a thorough postflight inspection is completed and treated with the same seriousness as

the preflight. Students often rush through postflight — set the standard early.

<u>Lesson Content (Ground)</u>

Taxiing Procedures

- Quiz the student on the before taxi check.
- Hand stays on the throttle.
- Heels on the floor (move feet up on the pedals when braking).
- Assist with wind corrections.
- Demonstrate with instrument checks.
- Assist with centerline and speed.
- Assist with runup.
- Ensure that every taxi is briefed <u>with an airport diagram</u> (except ramp).

Tracking Along a Road

- Review Chapter 7 (Ground Reference Maneuvers - Drift and Ground Track Control) of the Airplane Flying Handbook.
- Review how to track along a road with wind correction.

Straight and Level Flight Techniques

- Pitch + Power = Performance.
- Straight and level sight picture + 2300 RPM = 90KTS straight and level.
- Set pitch THEN power THEN trim.
- Utilize the HSI and inclinometer to maintain bank.
- Utilize the Altimeter and VSI to maintain altitude.

Go/No-Go Decision

- The student must make a go/no-go decision depending on:
  - Personal minimums.
  - FIT minimums.
  - Aircraft.
  - Airspace.
- Review the students planning with a focus on:
  - Weather, weight and balance, performance, flight planning, and fuel calculations.
  - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
  - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
  - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is

continuous and should be a consideration even once airborne.

Lesson Content (Dual)

Checklist Procedures

- Re-emphasize the importance of using checklists.
- Have the student run through all the checklists.
- Assist if needed.
- Have the student go through the “passenger briefing” and the “engine fire during start brief”.

Preflight Inspection

- Ensure that the student conducted a thorough preflight inspection.

Operation of Communications and Navigation Units

- Provide a brief introduction about COMS, NAV, GPS, audio panel, and transponder appropriate to the student Stage 1 knowledge.
  - Points of Emphasis:
    - Equipment ON/OFF
    - Database Currency
    - Volume
    - Squelch
    - Standby/Active Frequencies
    - Switch between navigation sources
    - Other useful functions such as: NRST page and Direct-To

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command’s responsibility to avoid traffic.
- Demonstrate how to scan for traffic in daytime conditions.
  - 10 degrees per second, looking both above and below aircraft level.
- IF YOU SEE SOMETHING, SAY SOMETHING

Taxiing Procedures

- Have the student perform the before taxi check.
- Hand stays on the throttle.
- Heels on the floor (move feet up on the pedals when braking).
- Assist with wind corrections.
- Demonstrate with instrument checks.
- Assist with centerline and speed.
- Assist with runup.

- Ensure that every taxi is briefed <u>with an airport diagram</u> (except ramp).

Review Effects of Controls as Appropriate
- If needed, refer to the lesson plan for 105.

Tracking a Ground Based Reference Line
- Use a straight-line ground reference and have the student track it.
- Ensure the student is correcting for wind drift.

Straight and Level Flight at Cruise Speed
- Demonstrate entering and maintaining straight and level flight.
- Pitch + Power = Performance.
- Set pitch THEN power THEN trim.
- Straight and level sight picture + 2300 RPM = 90KTS straight and level.
- Utilize the HSI and inclinometer to maintain pitch.
- Utilize the Altimeter and VSI to maintain altitude.

Straight and Level Flight at Selected Airspeeds
- For changes greater than 20 knots:
  - Full power or idle power.
  - Anticipate by 5 knots.
- For changes less than 20 knots:
  - 100 RPM change = 5 knots change

Straight and Level Flight with Flaps
- Demonstrate the effects of flaps 10, 25 and 40 degrees.
  Adding Flaps:
  - Lift → Increases.
  - Nose → Goes up.
  - Elevator (initially) → Forward.
  - Drag → Increases.
  - Airspeed → Decreases.
  - Elevator (due to drag) → Backward.

  Removing Flaps:
  - Lift → Decreases.
  - Nose → Goes down.
  - Elevator (initially) → Backward.

  - Drag → Decreases.
  - Airspeed → Increases.
  - Elevator (due to drag) → Forward.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 109.
- Assign homework:
  - Read PHAK chapter 7: Aircraft Systems

Completion Standards

This lesson is complete when the student is able to:

1. Have a weather briefing and a completed TOLD card prepared for the lesson, which will now become standard for all remaining flight lessons.
2. Demonstrate proper sequential checklist procedures for engine starting, before taxi, ground check, before takeoff, and runway entry.
3. Correctly perform previously learned maneuvers with minimal instructor guidance.
4. Demonstrate an understanding of straight and level flight techniques in the aircraft.
5. Explain the relationship between power, airspeed, and angle of attack at various speeds.